



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no. 2594102

Name of the Course: B.Tech.

Semester: II

Name of the Paper: Mathematics for AI-II

Paper Code: TMA-203

Time: 1.5 Hours

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks) (CO1)

a. Solve: $(e^y + 1) \cos x \, dx + e^y \sin x \, dy = 0$.

OR

b. Solve: $(x \sec^2 y - x^2 \cos y) \, dy = (\tan y - 3x^4) \, dx$.

Q2. (10 Marks) (CO1)

a. Solve: $\frac{dy}{dx} + \frac{y}{x} \log y = \frac{y}{x^2} (\log y)^2$

OR

b. Solve: $(x+1) \frac{dy}{dx} - 2y = (x+1)^4$

Q3. (10 Marks) (CO1)

a. Solve: $p^2 - 5p + 6 = 0$

OR

b. Solve: $x - y + p^2 = 0$

Q4. (10 Marks) (CO2)

a. Solve: $\frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} - 2y = e^x + \cos x$

OR

b. Solve: $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = \log x$.

Q5. (10 Marks) (CO2)

a. Solve: $\frac{d^2 y}{dx^2} + y = \sec x$ by using variation of parameters method.

OR

b. Solve: $(D^2 + 2D + 1)y = x \cos x$.